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RESEARCH ARTICLE

Variational Autoencoder with Latent Space Serialization as a Surrogate for Global Sea Surface Temperature

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Abstract

General Circulation Models (GCMs) or Earth System Models (ESMs), which provide projections of the Earth's future climate scenarios, rely on solving numerical equations for each grid to simulate the global climate data. This requires massive computational requirements, such as supercomputers, for each simulation. Moreover, GCMs exhibit biases relative to observed data with respect to various statistical measures. Recent advances in the domain of AI-based Weather Prediction (AIWP) address key bottlenecks of Numerical Weather Prediction (NWP) models by providing probabilistic, data-driven models that are also physics-inspired and can generate global/regional weather fields in a fraction of the time required by traditional simulations. However, AI-based weather prediction models are currently limited to weather scales and seasonal-to-annual timescales; generating physically consistent global climate fields over a large (e.g., 100-year) temporal scale still remains a major challenge. Recently, generative models have been applied to monthly global sea surface temperature (SST) maps, demonstrating skill at representing the spatial and temporal patterns of long-term climate variability. In this work, a Variational Autoencoder (VAE) with latent space serialization is proposed as a global SST emulator for a limited subset of Coupled Model Intercomparison Project Phase 6 (CMIP6) simulated data. Serialization of latent space enables the propagation of latent codes over time, thereby enhancing the capture of temporal variability. The model simulates global SST variability over 25-year horizons. It is trained on historical simulations of monthly SST data from the GISS-E2-1-H GCM model (r1i1p1f1), spanning 140 years (1850–1989), and the subsequent years are used for validation and testing. The model is rolled out for 200 years, and the results are evaluated with several experiments to understand the variability patterns. This work concludes with an analysis of results from multiple perspectives and outlines several avenues of future research that can leverage the potential of VAE latent space representation in capturing spatio-temporal variability in SST.

1. Introduction

GCMs or ESMs consider the physical, chemical, and biological processes to simulate the dynamics of the Earth system. They also consist of several coupled components, including the atmosphere, ocean, land surface, and sea ice (cryosphere), with ESMs additionally incorporating biogeochemical and chemical processes too. Climate variables that interact with ocean and atmosphere components, or atmosphere and land components, or any combination thereof, consist of several global and regional patterns of climate variability. One example is the Sea Surface Temperature (SST), a climate variable that interacts with almost all the components. Climate Surrogates are computationally efficient statistical or machine-learning models that are trained to approximate the outputs of GCMs or ESMs for climate variables. In recent years, AI-based weather prediction (AIWP) models (Bi *et al.*

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(2023),Kurth *et al.* (2023),Chen *et al.* (2023)) have emerged as lightweight probabilistic surrogates for numerical weather prediction. Most of the state-of-the-art AIWP models operate at daily to yearly timescales and remain limited in their ability to simulate long-term climate processes. Recent work has increasingly used Variational Autoencoders (VAEs) to compress high-dimensional climate fields into low-dimensional latent spaces that can be sampled to emulate physical variables, including sea surface temperature (SST). Recent developments in latent-space modeling for climate ensembles show that latent coordinates can be indexed by time, space, or an ensemble member. These coordinates can then be modeled via some regression or dynamical systems, effectively treating latent space as a serializable stochastic process. The present work explicitly positions a serialization-based VAE as a surrogate model for SST that encodes monthly SST maps into a VAE latent space and then propagates the latent codes over time using a Transformer model (Figure 1). The model is trained on simulated global SST anomaly data from the GISS-E2-1-H GCM (r1i1p1f1) for the period 1850-1989, and the results are validated and tested against data for 1991-2014. Several comparative studies have been conducted on the generated data for the validation years, extending up to 200 years to analyse long-term climate variability.

2. Literature Survey

Recently, generative models have shown promising results for emulating ocean variables, including conditional GANs (Generative Adversarial Networks) that replicate global SST maps from CMIP6 simulations, achieving strong spatial correlations and realistic ensemble variability (Chakraborty *et. al.* , 2025). Comparative analysis of VAEs, GANs, and diffusion models indicate that diffusion models more accurately capture fine-scale structures and variability for monthly SST, whereas GANs excel at representing distribution tails and extreme events. (Chakraborty *et. al.* , 2025). However, VAEs offer several advantages over GANs and diffusion models for SST emulation due to their explicit probabilistic latent structure and tractable likelihood-based training, which facilitate downstream temporal modeling of latent dynamics (Chakraborty *et. al.* , 2025). VAEs learn low-dimensional latent representations with explicit priors (typically Gaussian), enabling smooth interpolation, sampling, and latent traversals aligned with physically meaningful SST modes such as El Niño–Southern Oscillation (ENSO) variability and basin-scale trends (Solera-Rico *et. al.* , 2024). VAE is employed to nonlinearly decompose monthly SST anomalies into a low-dimensional latent space, enabling the generation of realistic SST anomaly fields from this representation (Schlör and Goswami , 2021). In contrast, GANs can generate high-quality samples but lack a structured and interpretable latent space, limiting their suitability for controlled SST generation or latent dynamical modeling (Besombes *et. al.* , 2021). Diffusion models operate in high-dimensional state space or compressed latents (e.g., via VAEs), but their representations are primarily optimized for denoising rather than explicit probabilistic structure, making VAE latent spaces more amenable to physics-informed constraints and hybrid dynamical frameworks (Meuer *et. al.* , 2025).

VAEs effectively reproduce high-resolution spatial extremes (e.g., SST maxima (Zhang *et al.* , 2023), (Ma *et al.* , 2025)) at low post-training computational cost and outperform EOF-based methods when applied to nonstationary and non-Gaussian climate data. Vanilla convolutional VAEs have been used to compress large climate datasets (e.g., vertical velocity, spatial priors) while preserving variance and structure, ideal for SST fields where spatial correlation dominates (Semenova *et al.* , 2022). VAEs are optimized under a maximum-likelihood objective that balances reconstruction fidelity and latent regularization. Moreover, their structured latent spaces are well-suited for coupling with temporal models such as Long Short-Term Memory (LSTMs) or Transformers for auto-regressive SST emulation (Paçal *et. al.* , 2025). A hybrid model for SST prediction combines ConvLSTM-style networks with Gaussian-process components to capture both spatial and temporal dependencies of SST (Beauchamp et al. , 2025). Recent advances in physics-informed loss functions, conditioning generative models on external forcings, and hybrid architectures with explicit dynamical components point toward the next

generation of emulators that better balance realism, stability, and physics interpretability (Meuer *et. al.* , 2025).

Recent work on VAEs for SST emulation has focused increasingly on using the latent space as a compact surrogate for high-dimensional SST fields, but explicit serialization of that latent space over time for emulation remains a nascent direction. VAE latent representations can be combined with temporal models e.g., autoregressive or Transformer-based dynamics to capture the evolution of latent codes. The latent space of a VAE can effectively compress complex, high-resolution climate data into a lower-order manifold that remains physically interpretable (Behrens et al. , 2022). In atmospheric and oceanic contexts, these latent nodes are capable of differentiating between different physical regimes such as convective modes or circulation patterns more effectively than conventional methods like EOF (Spuler *et. al.* , 2025). The VAE latent space serialization approach as a surrogate for global SST, as employed in this work, is the first of its kind effort in this direction.

The following section describes the r1i1p1f1 ensemble data of the GISS-E2-1-H model and the associated data preprocessing steps. Section 4 presents the methodology, including the VAE architecture, the training process, the emulation of SST anomaly fields, and the evaluation metrics used to assess the reconstruction performance. Section 5 reports the experimental results, with a discussion of global and regional modes of variability observed, and the bias during the rollouts. Section 6 concludes the study.

3. Dataset and Pre-processing

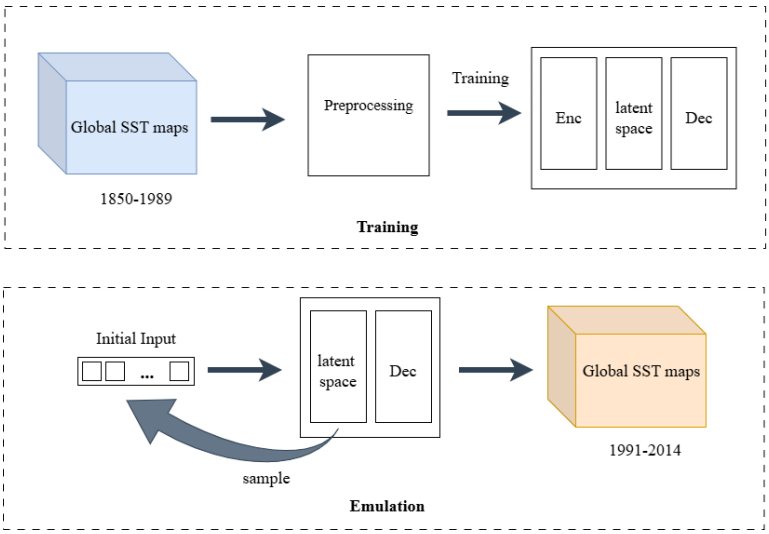

Figure 1. *Workflow of training and emulation of the proposed surrogate model.*

3.1. Description of Dataset

To develop the surrogate generative model, we employed historical SST simulations from GCM models participating in the CMIP6 project. In particular, we used the r1i1p1f1 ensemble data of the GISS-E2-1-H model (Tatebe *et. al.* (2019)). The dataset consists of monthly SST fields for the period of 1850–2014 at a spatial resolution of $1^\circ \times 1^\circ$. To compare the results with respect to the observed data, global reanalysis dataset from the Centennial in situ Observation-Based Estimates (COBE) 2 dataset (Schneider *et. al.* (2013)) is considered. This dataset contains monthly global SST data at a resolution of $1^\circ \times 1^\circ$ (1891 to 2023).

3.2. Preprocessing of Dataset

The collected datasets undergo several preprocessing steps to be suitable for CNN input. Due to limited GPU availability, the experiments are conducted on a coarser scale (3.75° × 2.5°) of SST maps. It resulted in a map shape of (48 × 144), which preserves most of the important variations of the full-resolution dataset.

As the global SST data is dominated by the linear trend and annual pattern, and the CNN models are prone to capture the mean pattern, the datasets further have to undergo anomaly calculation (Lau (1997)) and removal of linear trend (Hess *et. al.* (2001)). Finally, the datasets are normalised to adjust the data to a normal distribution. The final simulated dataset is shaped as (165 (y) × 12 (m) × 48 (lat) × 144 (lon)), from which the first 140 years (1850 - 1989) are considered for training, the next 14 years are considered for validation (with 1 year buffer with respect to train set), and the next 10 years for testing.

4. Methodology

This study employs a sequential variational autoencoder (VAE) with a Transformer model as a learned conditional prior to model the spatiotemporal variability of SST anomalies derived from a CMIP6 GCM model ensemble. Unlike standard VAEs, which assume an independent and identically distributed (i.i.d.) latent prior, the proposed framework explicitly models temporal dependencies in the latent space via a history-conditioned prior.

The model is trained to learn a probabilistic, low-dimensional representation of SST anomaly fields while capturing their temporal evolution. The overall methodological framework consists of four components: (i) a convolutional encoder–decoder architecture, (ii) a Transformer-based prior network for temporal latent dynamics, (iii) variational inference for joint learning of posterior and prior distributions, and (iv) autoregressive generation of SST anomaly sequences.

4.1. Serialized VAE Architecture

The model operates on two-dimensional gridded SST anomaly fields of size 48 × 144, where each monthly anomaly map is treated as a single-channel input. The architecture consists of three main components: an encoder, a conditional prior network, and a decoder (Figure 2).

Encoder. The encoder $q_\phi(\mathbf{z}_t|\mathbf{x}_t)$ is implemented as a convolutional neural network that maps each SST anomaly field $\mathbf{x}_t \in \mathbb{R}^{1\times 48\times 144}$ to a latent representation. It consists of three convolutional layers with kernel size 4 × 4 and stride 2, progressively reducing spatial resolution while increasing channel depth (64, 128, and 256 channels). Each convolutional layer is followed by batch normalization and ReLU activation.

The resulting feature map of size 256×6×18 is flattened and passed through a fully connected layer to produce the parameters of a Gaussian posterior distribution:

$$q_\phi(\mathbf{z}_t|\mathbf{x}_t) = \mathcal{N}(\boldsymbol{\mu}_t^{(q)}, \mathrm{diag}(\boldsymbol{\sigma}_t^{2(q)})), \tag{1}$$

where $\mathbf{z}_t \in \mathbb{R}^D$ is the latent variable.

Conditional Prior Network. Unlike standard VAEs, the prior is conditioned on both past latent states and previous observations:

$$p_\theta(\mathbf{z}_t|\mathbf{z}_{t-H:t-1}, \mathbf{x}_{t-1}). \tag{2}$$

The latent history $\mathbf{z}_{t-H:t-1}$ is processed using a Transformer encoder with sinusoidal positional encoding to capture temporal dependencies. The Transformer outputs a context-aware latent representation corresponding to the most recent timestep.

In parallel, the previous SST field $\mathbf{x}_{t-1}$ is processed using a convolutional network to extract spatial features. These features are concatenated with the Transformer output and passed through fully connected layers to produce the parameters of the conditional prior:

$$p_\theta(\mathbf{z}_t|\cdot) = \mathcal{N}(\boldsymbol{\mu}_t^{(p)}, \mathrm{diag}(\boldsymbol{\sigma}_t^{2(p)})). \tag{3}$$

Decoder. The decoder $p_\theta(\mathbf{x}_t|\mathbf{z}_t)$ reconstructs SST anomaly fields from latent variables. The latent vector is first mapped to a high-dimensional feature space via a fully connected layer and reshaped to $256 \times 6 \times 18$. A sequence of transposed convolutional (deconvolutional) layers progressively upsample the feature maps back to the original resolution. Batch normalization and ReLU activations are applied in intermediate layers, and the final layer outputs a single-channel reconstruction:

$$\hat{\mathbf{x}}_t = p_\theta(\mathbf{x}_t|\mathbf{z}_t). \tag{4}$$

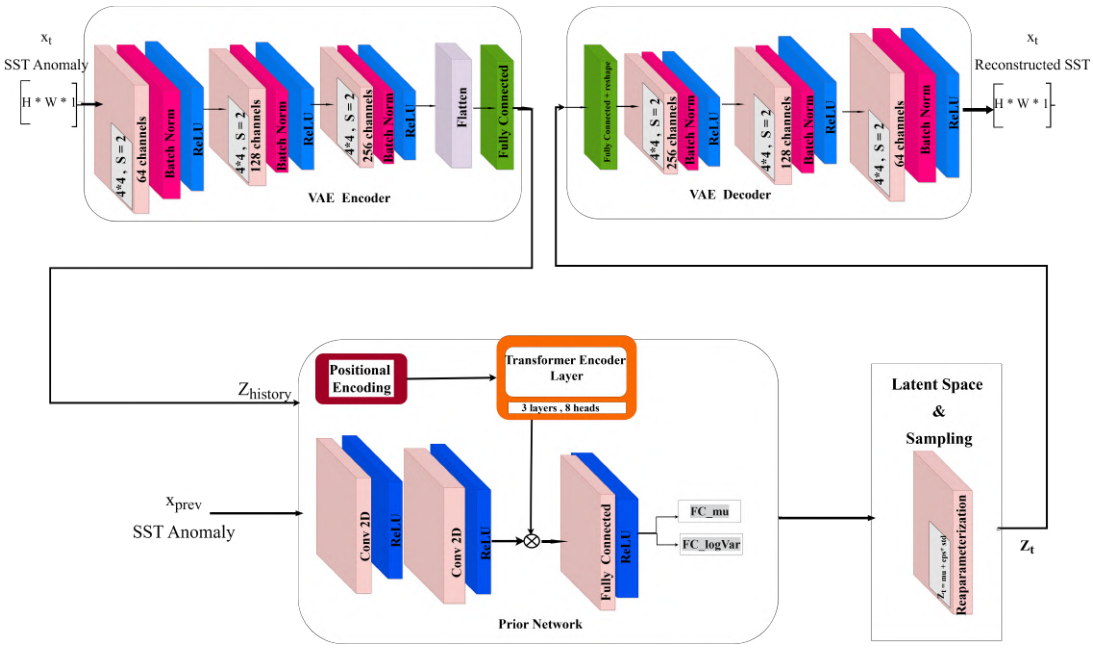

Figure 2. *Schematic Diagram of the Serialized Variational Auto Encoder: A Transformer model is incorporated as a learned conditional prior to model the spatiotemporal variability .*

4.2. Training

Prior to training, SST fields are preprocessed to isolate internal climate variability by removing the long-term climatological mean and linear temporal trends. Monthly SST anomalies are computed relative to the climatological mean, and missing values are set to zero to ensure computational consistency during training.

Training Objective. The model is trained by maximizing a time-dependent evidence lower bound (ELBO), which balances reconstruction accuracy and consistency between posterior and prior distributions:

$$\mathcal{L} = \mathbb{E}_{q_\phi(\mathbf{z}_t|\mathbf{x}_t)}[\log p_\theta(\mathbf{x}_t|\mathbf{z}_t)] - \beta\,\mathrm{KL}\big(q_\phi(\mathbf{z}_t|\mathbf{x}_t) \,\|\, p_\theta(\mathbf{z}_t|\mathbf{z}_{t-H:t-1}, \mathbf{x}_{t-1})\big). \tag{5}$$

The KL divergence is computed between two Gaussian distributions.

The dataset is split along the temporal dimension into training, validation, and test subsets. Training is performed using mini-batches, and the model achieving the lowest validation loss is selected for subsequent analysis. The optimization details of the training process are provided in Supplementary Section A.

4.3. Autoregressive Emulation of SST Anomalies

After training, the model is used as a stochastic emulator to generate SST anomaly sequences in an autoregressive manner. Unlike standard VAEs, the proposed model generates sequences by recursively sampling from the learned conditional prior.

At each timestep, the latent variable is sampled as:

$$\mathbf{z}_t \sim p_\theta(\mathbf{z}_t | \mathbf{z}_{t-H:t-1}, \mathbf{x}_{t-1}), \tag{6}$$

and the corresponding SST anomaly field is obtained via the decoder:

$$\hat{\mathbf{x}}_t = p_\theta(\mathbf{x}_t | \mathbf{z}_t). \tag{7}$$

The generated output $\hat{\mathbf{x}}_t$ is then fed back as input for the next timestep, forming an autoregressive generation process. This procedure enables the model to simulate temporally coherent SST anomaly sequences consistent with the learned latent dynamics.

4.4. Evaluation Metrics

Model performance is evaluated using both reconstruction-based and dynamical metrics. Reconstruction accuracy is quantified using the mean squared error (MSE):

$$\mathcal{L}_{\text{rec}} = \mathbb{E}\left[\|\mathbf{x}_t - \hat{\mathbf{x}}_t\|_2^2\right], \tag{8}$$

where $\hat{\mathbf{x}}_t = p_\theta(\mathbf{x}_t | \mathbf{z}_t)$ is the reconstructed SST anomaly field. In addition, dynamical consistency is assessed by comparing spatial variance, temporal mean structure, and spectral characteristics between the generated and reference SST fields using mean, standard deviation (std), Empirical Orthogonal Function (EOF), and Power Spectrum Analysis (PSA).

5. Experimental Results

The experiment was carried out on a dataset of the GISS-E2-1-H model. The data were smoothed by calculating a 3-month moving average window temporally to remove the random noise. The analysis is divided into two parts: analysing the global spatiotemporal mode of climate variability and the regional modes of climate variability.

5.1. Global Modes of Climate Variability

Several phenomena in the climate affect SST variability, leading to a spatiotemporal distribution of variations. Examining the variability patterns and their amplitude reveals several climatic phenomena on SST. The mean, standard deviation, and Empirical Orthogonal Function (EOF) patterns of the emulated data are compared with those of the GISS model data to assess whether the proposed emulator captures the climate's physics from the data itself.

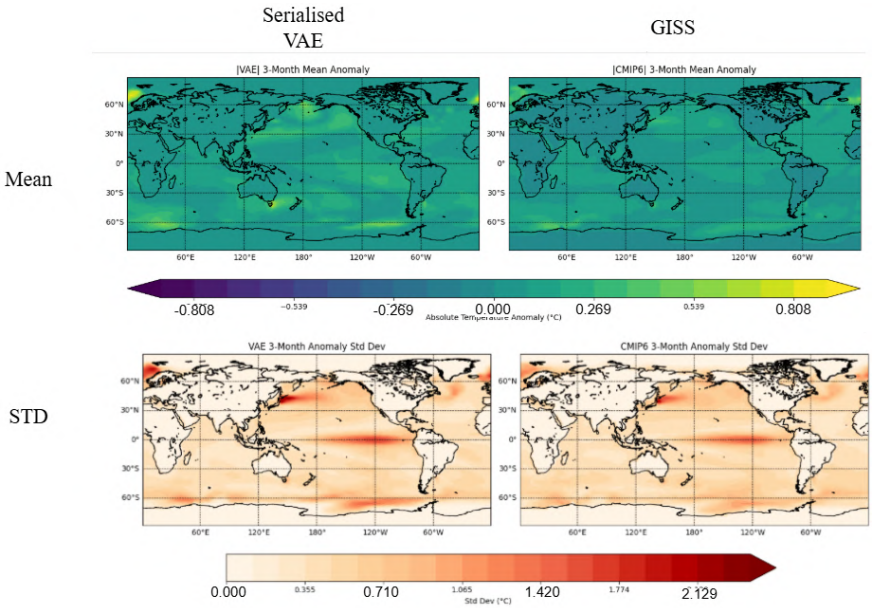

Figure 3. *Mean and Standard Deviation of the serialized VAE and GISS model data. The serialized VAE captures the large-scale spatial structure of variability, including the enhanced variance in the tropical Pacific and weaker variability in subtropical regions. The temporal-mean maps are broadly consistent between the two datasets .*

5.1.1. Mean and Standard Deviation

The heatmap of the mean and standard deviation of the global SST anomaly highlights the global variability pattern. In this analysis, the temporal mean and standard deviation statistics of the dataset generated by the serialized VAE are compared with the GISS data on which it was trained.

In figure 3, the standard deviation (std) and mean map of the serialized VAE are compared with the simulated dataset (GISS-E2-1-H). The std maps show that the serialized VAE captures the large-scale spatial structure of variability, including the enhanced variance in the tropical Pacific and weaker variability in subtropical regions. However, the generated field has a higher variance amplitude than GISS-E2-1-H, with greater overestimation in regions of intense variability (notably the equatorial Pacific). The temporal-mean maps are broadly consistent between the two datasets: both are centered near zero and reproduce similar spatial patterns. However, the VAE mean field exhibits sharper spatial gradients and more pronounced local extrema, suggesting reduced spatial smoothness relative to the reference dataset.

Overall, these diagnostics indicate that the serialized VAE learns the first-order and large-scale second-order statistics of SST anomalies, but it overestimates the magnitude of variance, hence amplifying the fine-scale spatial structure. Therefore, while the model is suitable as a baseline generative emulator, improving variability calibration and spatial smoothness may require additional regularization or increased training data. Further information on the global patterns recognised by the proposed model is discussed with the results of the EOF analysis.

5.1.2. EOF Analysis

EOF analysis is closely related to PCA, but is specifically applied to two-dimensional spatial fields. This technique decomposes the spatiotemporal variability of the data into a set of orthogonal spatial patterns (Empirical Orthogonal Function, EOF) and their corresponding temporal distributions of mean amplitudes for each Principal Component (PC). This temporal distribution also captures the short- and

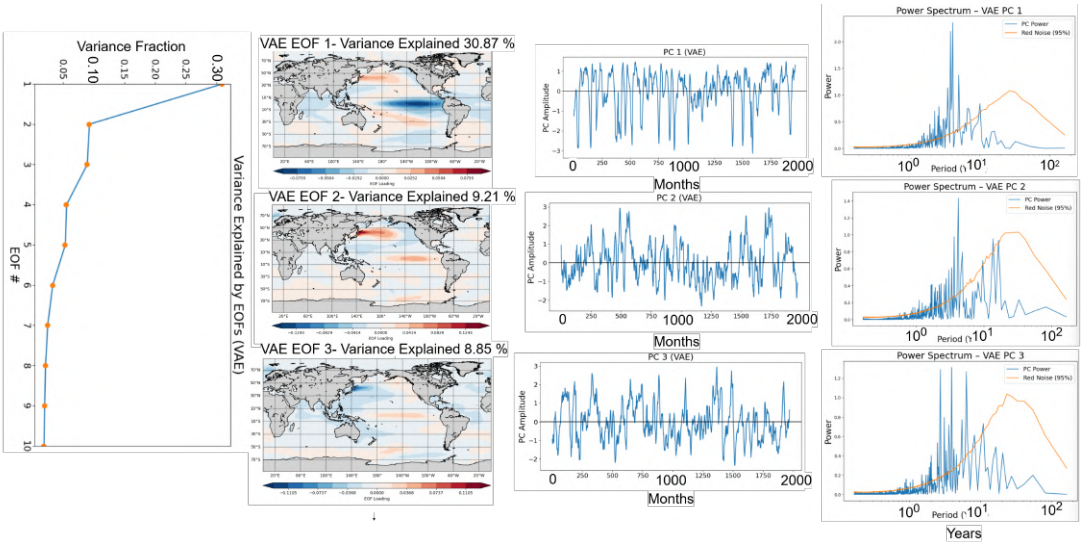

Figure 4. *EOF analysis of the first 3 components of the data generated by Serialized VAE data for test period. The first three PCs of the GISS data account for about 50% of the total variance.*

long-term oscillation frequencies of each component, as shown in the Power Spectrum Analysis (PSA) for each component.

In this section, we intend to compare the spatial and temporal patterns of the EOFs of the serialized VAE (Figure 4) with the simulated data by CMIP6 (GISS) (Supplementary figure 3). The first three PCs of the GISS data account for about 50% of the total variance. The spatial map of the first EOF of GISS shows the ENSO signal as the dominant mode, accounting for 24.46% of the variance. The corresponding amplitude series and PSA show two peaks within a 10-year period that cross the Gaussian noise limit, indicating oscillations like ENSO. In the case of VAE, the first PC accounts for a similar fraction (30%) of the total variance, and the spatial map shows that most of the variability is due to the ENSO pattern. The spatial structure of the first EOF exhibits a clear equatorial Pacific pattern, consistent with the canonical ENSO signal. This demonstrates that the model successfully captures the dominant large-scale mode of SST variability. The second EOF pattern shows the distribution over north pacific ocean mostly, and some residual of ENSO. The PSA shows some significant peaks after 10 years, which denote the decadal oscillation (e.g., PDO). Finally the third EOF shows the residual of the previous oscillations. On the other hand, the EOF patterns of the GISS model data also contain similar spatial and temporal patterns, but with the EOF2 and EOF3 exchanged, with a negligible difference in variance fraction. The model's 200-year rollout (Supplementary Figure 7) demonstrates its consistency in capturing the dominant modes of climate variability. But the power of the decadal oscillation in long-term simulation is slightly reduced.

The results conclude that the serialized VAE can identify the spatial and temporal pattern of climate variability more effectively. The key point is that temporal coherence is improved by serialization; further improvements can be achieved by introducing more GCM models for training data and fine-tuning with observed data. This motivates future work in designing a more extensive training process.

5.2. Regional Mode of Climate Variability: ENSO

Apart from the aforementioned global modes, the SST contains various regional oscillations of seasonal to multidecadal time scales. In the following experiment, the simulation of a well-known regional phenomenon, ENSO, a seasonal oscillation over the eastern tropical Pacific Ocean, is considered. ENSO

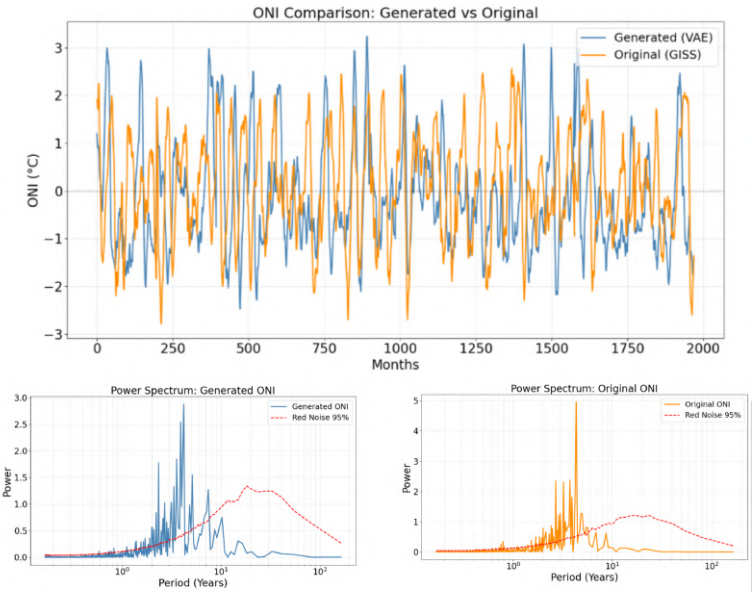

Figure 5. *Oceanic Nino Index (ONI) pattern and Power Spectrum Analysis for CMIP6 (GISS-E2-1-H r1i1p1f1) and serialized VAE generated data. Several prominent peaks in the generated series align with those in the reference data, suggesting that the model captures large-scale ENSO events (El Niño and La Niña).*

plays a fundamental role in modulating the global climate system. ENSO is characterised by alternating warm and cold conditions in the central and eastern equatorial Pacific Ocean, which influence the Rossby waves and, in turn, affect global rainfall patterns. ENSO variability is commonly quantified using the Oceanic Niño Index (ONI), defined as the three-month running mean of sea surface temperature anomalies averaged over the Niño 3.4 region (5°S–5°N, 170°W–120°W).

Figure 5 presents a comparison between the ONI time series generated by the Serialized VAE and the corresponding CMIP6 (GISS-E2-1-H) test data.

The variability of the generated ONI is comparable to the GISS model data, indicating that the model is able to reproduce the magnitude of SST anomalies more effectively. Additionally, several prominent peaks in the generated series align with those in the reference data, suggesting that the model captures large-scale ENSO events (El Niño and La Niña). Further insight is obtained from the Power Spectrum Analysis (PSA). The spectrum of the original GISS data shows a prominent peak in the seasonal band (approximately 2–7 years), consistent with the known periodicity of ENSO. In comparison, the Serialized VAE also exhibits increased spectral power within this range, indicating a fair representation of ENSO-related timescales. However, the generated spectrum remains broader and noisier, with energy spread over a wider frequency range. Overall, the Serialized VAE effectively captures both the amplitude and temporal variability of ENSO. The incorporation of temporal conditioning through latent serialization enables better alignment with the climate dynamics.

Apart from ENSO, the Indian Ocean Dipole (IOD) and Atlantic Multi-decadal Oscillation (AMO) are also calculated and compared with GISS. The time series plot and PSA of IODM and AMO index are shown in Supplementary Figure 10 and 9. The results conclude that the seasonal oscillation of IOD and the multidecadal oscillation of AMO are perfectly captured in the proposed model. It shows that the proposed emulator model can recognise the physics of the climate dynamics from the training data.

5.3. Mean and Std Bias

The temporal bias of CMIP6 models relative to observed data is well established. Nonetheless, bias correction of the model output is outside the scope of this research work. Instead, this work focuses on ensuring that the VAE-generated data does not amplify bias to a greater extent than the training data. The Mean and standard deviation bias maps with respect to COBE observations (1991–2014) for both the GISS data and serialized VAE outputs provide an analytical assessment of model bias (Figure 8).

The GISS model data shows an average mean and std bias of 0.085 and 0.568, whereas the serialized VAE shows average bias values of 0.112 and 0.590. Moreover, the spatial patterns of the bias are also similar. The similarity in bias patterns and average bias values of the data indicates that the serialized VAE inherits the bias of the training data. Further increasing the diversity of the training data and fine-tuning with observed data can reduce bias. This motivates the future work of this paper.

6. Conclusion

This research concludes with key findings from statistical analyses of the serialized VAE as a surrogate model for simulated SST data. The standard deviation, EOF, PSA, etc., conclude that the Serialized VAE captures the spatio-temporal pattern of the data very well but fails to meet the optimal temporal standard deviation of the data. These may be caused by a lack of variation in the training data. The proposed model is able to recognize the regional phenomenon (e.g., ENSO, AMO, etc.) and their oscillation patterns (from seasonal to multidecadal scale), which denotes that the proposed model could learn the physical properties of climate from the data itself.

The above points motivate a larger scope of future work for this research. Expanding the training data of the base model and introducing observed data for finer-tuning to include a wider variety of datasets would enable it to learn a broad spectrum of SST variability and standard deviation. Additionally, incorporating anthropogenic forcings and multiple climate variables could increase the model's information content and support projections of future climate scenarios. The VAE framework also enables the storage of essential features within a low-dimensional latent space, from which additional information can be decoded as required. This approach, known as Omnicast, reduces computational costs and improves the portability of the methodology. As such, it aligns closely with the methodology of this study and represents a promising direction for future research.

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Competing Interests. The authors declare no competing interests exist.

Data Availability Statement. The CMIP6 and COBE data have been downloaded from the following links.
CMIP6: https://www.wdc-climate.de/ui/cmip6?input=CMIP6.CMIP.NASA-GISS.GISS-E2-1-H
COBE: https://psl.noaa.gov/data/gridded/data.cobe.html
The preprocessed data used for the paper is uploaded along with the code in Zenodo.

Code Availability Statement. The reproducible code, dataset, and final weights are uploaded through the following link. https://zenodo.org/records/19352481

Author Contributions. The first author has done the coding experiments mainly and contributed to the draft. The second author has drafted the paper and helped to maintain its integrity. The third author has formulated the problem and refined the paper, and the fourth author has supervised the project.

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Supplementary:

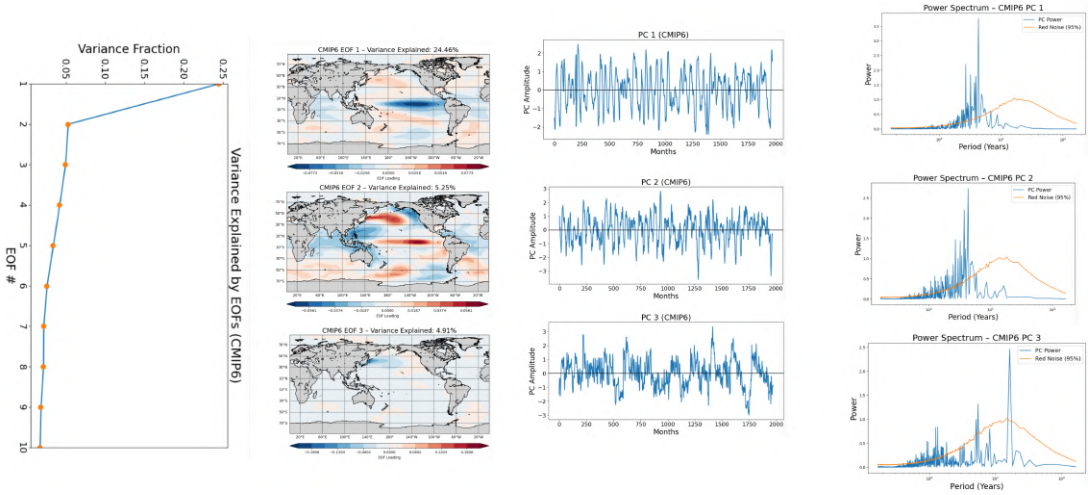

Figure 6. *EOF analysis of the first 3 components of the CMIP6 (GISS-E2-H r1i1p1f1) for test period. .*

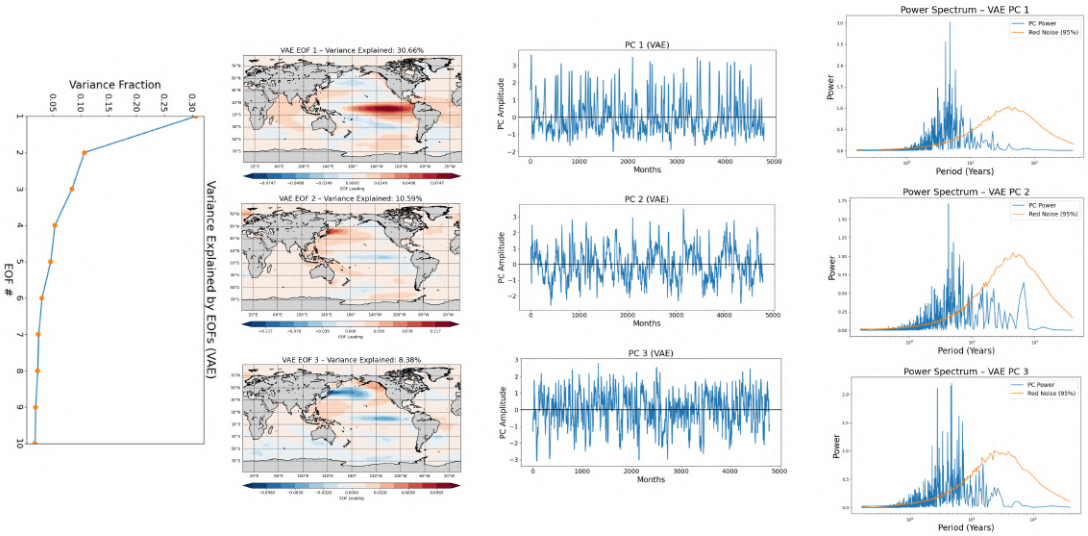

Figure 7. *EOF analysis of the Serialised VAE generated data for 200 years of roll over, shows consistency in capturing the dominant modes of climate variability, but the power of the decadal oscillation in long-term simulation is slightly reduced .*

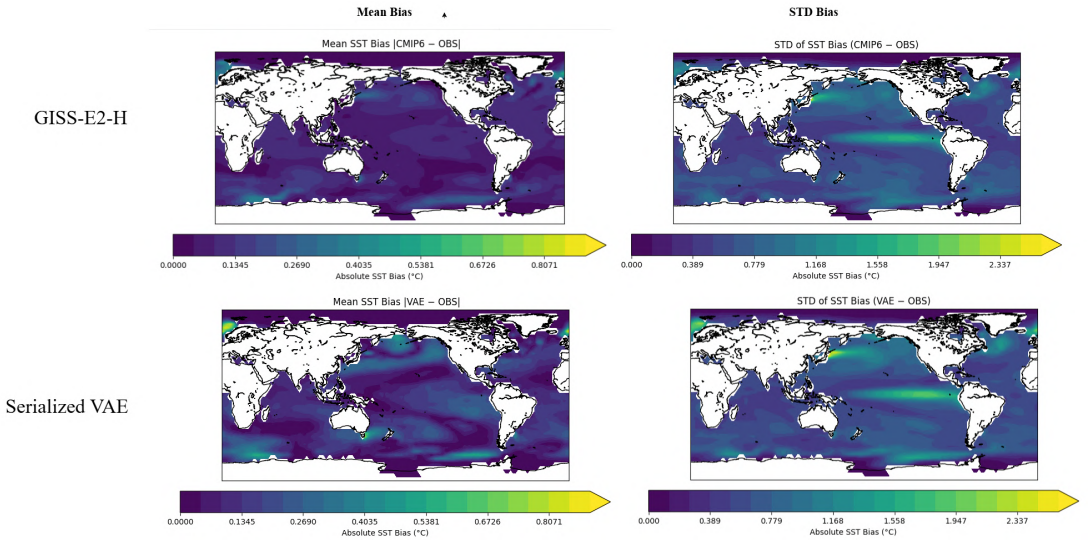

Figure 8. *Mean and Standard Deviation bias for CMIP6(GISS-E2-H r1i1p1f1) and Serialised VAE generated data..*

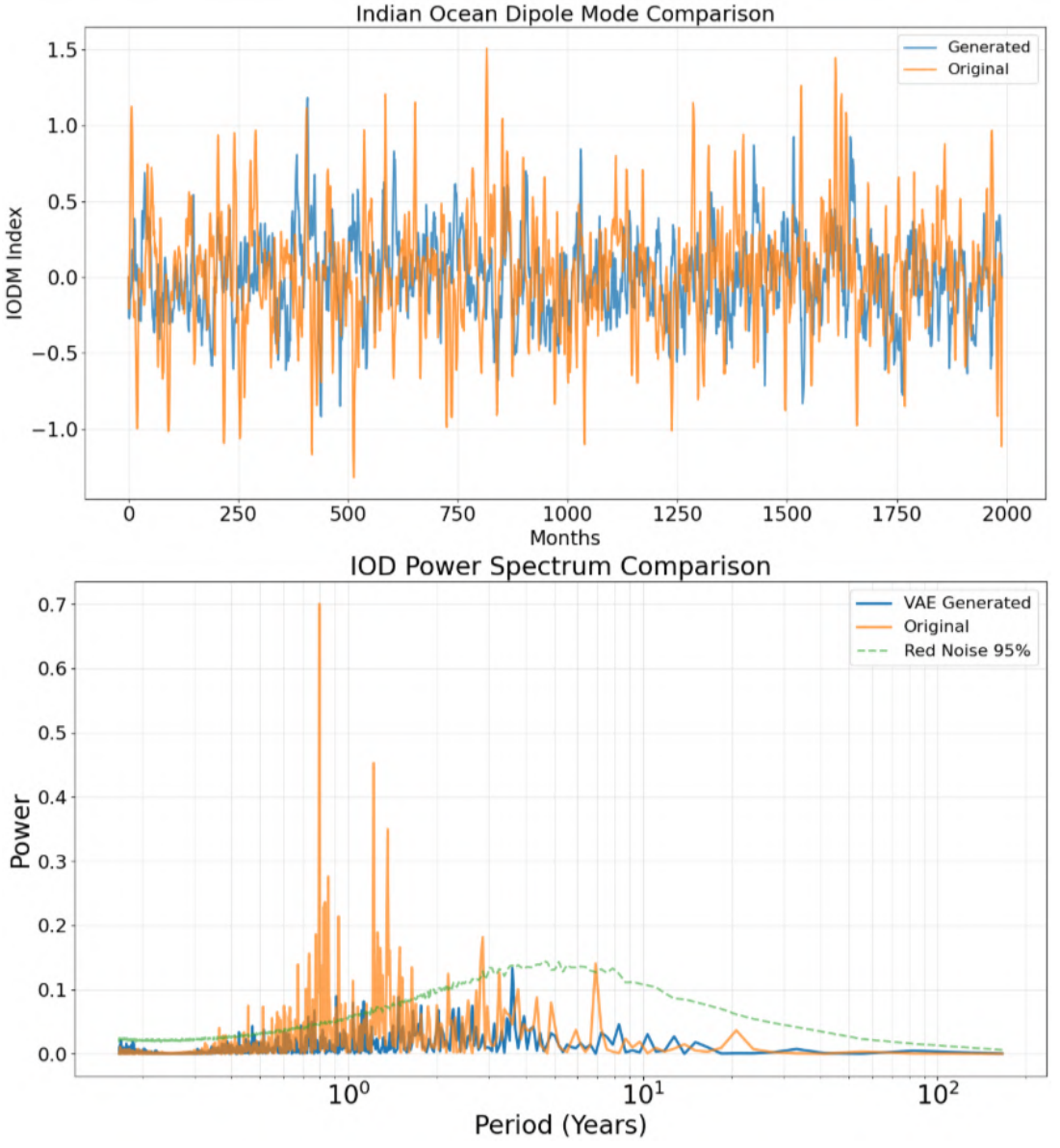

Figure 9. *IOD Index and PS Comparison for CMIP6(GISS-E2-H r1i1p1f1) and Serialised VAE generated data.*

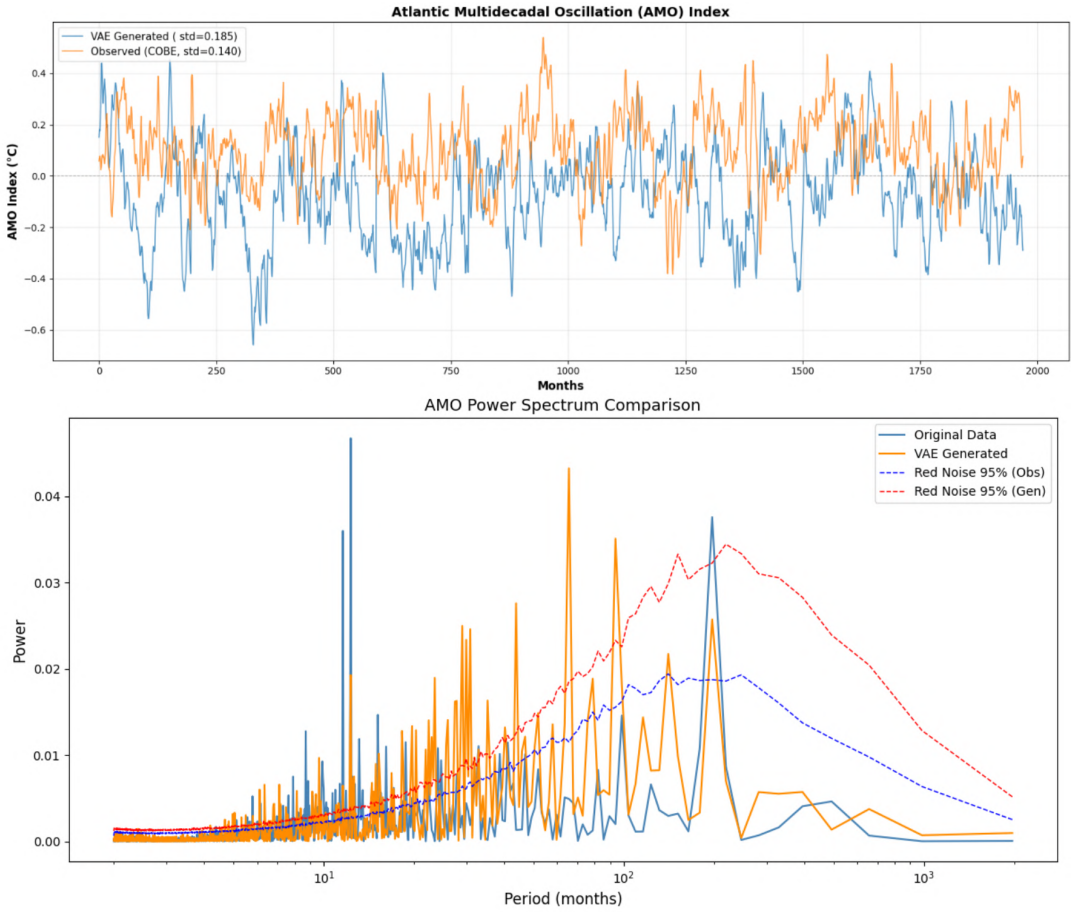

Figure 10. *AMO Index and PS Comparison for CMIP6(GISS-E2-H r1i1p1f1) and Serialised VAE generated data..*

A. Training Objective and Optimization Details

The model is trained by maximizing a time-dependent evidence lower bound (ELBO), which balances reconstruction accuracy and consistency between the approximate posterior and a learned conditional prior:

$$\mathcal{L} = \mathbb{E}_{q_\phi(\mathbf{z}_t|\mathbf{x}_t)}\left[\log p_\theta(\mathbf{x}_t|\mathbf{z}_t)\right] - \beta\,\mathrm{KL}\left(q_\phi(\mathbf{z}_t|\mathbf{x}_t) \parallel p_\theta(\mathbf{z}_t|\mathbf{z}_{t-H:t-1}, \mathbf{x}_{t-1})\right). \tag{9}$$

Unlike standard VAEs, the prior distribution is not fixed but is parameterized by a neural network conditioned on past latent states and previous observations.

Reconstruction Loss.. The reconstruction term is implemented using a mean squared error (MSE) loss:

$$\mathcal{L}_{\mathrm{rec}} = \mathbb{E}\left[\|\mathbf{x}_t - \hat{\mathbf{x}}_t\|_2^2\right], \tag{10}$$

where $\hat{\mathbf{x}}_t = p_\theta(\mathbf{x}_t|\mathbf{z}_t)$ is the reconstructed SST anomaly field.

KL Divergence.. The Kullback–Leibler divergence is computed between two multivariate diagonal Gaussian distributions:

$$q_\phi(\mathbf{z}_t|\mathbf{x}_t) = \mathcal{N}(\boldsymbol{\mu}_t^{(q)}, \mathrm{diag}(\boldsymbol{\sigma}_t^{2(q)})), \quad p_\theta(\mathbf{z}_t|\cdot) = \mathcal{N}(\boldsymbol{\mu}_t^{(p)}, \mathrm{diag}(\boldsymbol{\sigma}_t^{2(p)})).$$

The KL divergence has the closed-form expression:

$$\mathcal{L}_{\mathrm{KL}} = \frac{1}{2}\sum_{i=1}^{D}\left[\log\frac{\sigma_{t,i}^{2(p)}}{\sigma_{t,i}^{2(q)}} + \frac{\sigma_{t,i}^{2(q)} + (\mu_{t,i}^{(q)} - \mu_{t,i}^{(p)})^2}{\sigma_{t,i}^{2(p)}} - 1\right]. \tag{11}$$

Final Training Objective.. In practice, the model is trained by minimizing the negative ELBO:

$$\mathcal{L} = \mathcal{L}_{\mathrm{rec}} + \beta\,\mathcal{L}_{\mathrm{KL}}, \tag{12}$$

where β is a weighting coefficient controlling the strength of regularization. In this study, $\beta = 0.01$.